

ROBERT GABRIEL MACALINTAL

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EXPERIENCE

Bioinformatics Software Developer

Sept 2024 – Aug 2025

BICoE Team – Agriculture and Agri-Food Canada, Ottawa, ON

<https://github.com/AAFC-BICoE/dina-py> & <https://github.com/AAFC-BICoE/dina-local-deployment>

- Worked on the DINA project – a FOSS project / web application collaborated on by multiple government departments such as AAFC, DFO, and Health Canada – React frontend, Postgres database, Java backend
- Used Python and ORM concepts and methodologies to migrate data from a legacy web application to a newer application which improved data-entry efficiency by 100%
- Developed Python migration scripts based on data given by other departments to migrate Excel sheet data into the DINA web application
- Updated Helm chart to fix migration problems that was found in RabbitMQ
- Migrated 10+ Enzyme tests to React Testing Library tests (Frontend tests)
- Updated tickets to document and keep track of work item progress

Help Desk Analyst

May 2026 – Present

National Capital FreeNet, Ottawa, ON

- Managed and configured assets like modems and modem/router combos
- Updated existing documentation and tickets to help clients with network issues

IT Technician

Jan 2024 – Apr 2024

NITCSS Team – Agriculture and Agri-Food Canada, Ottawa, ON

- Evergreen Project – Recycled old computers and set them up for donation or e-waste
- Troubleshooted and used automated scripts through MECM to fix client software issues remotely

PROJECTS

Forgeborn – A D&D companion web application

<https://github.com/rgmacalintal/project-hexblade> - all work progress made in the 'dev' branch

- MVC Architecture
- ASP.NET Core / C# backend component
- React.js frontend component
- MySQL database

Battleship – Recreation of the game Battleship

<https://github.com/rgmacalintal/Battleship>

- Java 17 backend, and Java Swing frontend
- Dev mode enabling developers to see the computer opponent's board
- Some internationalization implementations to switch between English and Tagalog
- Network component enabling players to play against each other (dev mode disabled)

Pong – Recreation of the classic Pong Game

<https://github.com/rgmacalintal/Pong>

- Build in Unreal Engine 5 and coded in C++
- Implemented paddle movement, ball physics, and collision detection systems
- Designed simple AI logic for opponent paddle behavior

SKILLS

Programming Languages: Bash, C, C#, C++, CSS, HTML, Java, JavaScript, JSON, PHP, PowerShell, Python, R, SQL, TypeScript, XML, YAML/YML

Frameworks: ASP.NET Core, Entity Framework Core, Node.js

Tools and Technologies: Azure, Docker, Eclipse, Git, GitHub, Oracle VirtualBox, Postman, React Testing Library, Selenium IDE, Swing, Unix / Linux, VSCode, Visual Studio, VMWare

Databases: MariaDB, Microsoft SQL Server, MySQL, Oracle, PostgreSQL, SQLite

Concepts: Agile, Bitwise Operations, Compilation Process, CI/CD, Data Migration, Data Structures, Debugging, IPC / Concurrent Programming, Mathematical Algorithms, MVC Architecture, Networking Fundamentals, OOP, ORM, Real-time Programming, RESTful APIs, SDLC, SOLID, Testing, Troubleshooting, WCAG

EDUCATION

Computer Engineering Technology - Computing Science Advanced Diploma	2023 – 2025
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Algonquin College, Ottawa, ON

- Graduated with Honours
- Co-operative Education

Bachelor of Science with Major in Mathematics	2015 – 2020
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University of Ottawa, Ottawa, ON

REFERENCES

Available upon request.